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 / **Practice Exam - 1**

Correct Answer

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Practice Exam - 1

Question 47 of 117



A 38-year-old previously healthy woman has been suffering from slowly progressive exertional dyspnea over the last 3 years. Her medication history is notable for remote use of prescription diet pills. Pulmonary function tests reveal a 20% drop in DLCO. Echocardiogram reveals normal left ventricular size and function, a dilated pulmonary artery, right ventricular and right atrial dilation, and moderate tricuspid regurgitation with an estimated RV systolic pressure 70 mm Hg. She undergoes a right heart catheterization which reveals:

- RA 8 mm Hg
- PA 76/40 (mean 53 mm Hg)
- PCW 11 mm Hg
- CO 5.0 L/min
- PVR 8.4 Wood units

Which of the following would constitute a positive vasodilator challenge that would prompt initiation of long-term calcium channel blocker therapy?



A. After 5 minutes of 40 p.p.m. inhaled NO, mPAP falls to 25 mm Hg and CO remains at 5.0 L/min



B. After 5 minutes of 40 p.p.m. inhaled NO, mPAP falls to 41 mm Hg and CO rises 10% to 5.6 L/min

C. After 5 minutes of 40 p.p.m. inhaled NO, mPAP falls to 25 mm Hg and CO decreases to 3.5 L/min

D. Vasodilator challenge is not indicated.

Commentary

References

Testing point: The Board-certified AHFTC specialist should recognize the indications for vasodilator therapy in pulmonary arterial hypertension patients.

Answer: A. After 5 minutes of 40 p.p.m. inhaled NO, mPAP falls to 25 mm Hg and CO remains at 5.0 L/min.

This patient has anorexigen-associated pulmonary arterial hypertension. The original study revealing pulmonary vasoreactivity as a predictor of calcium-channel blocker responsiveness was done in idiopathic PAH subjects.¹ A positive response to pulmonary vasodilator testing should include all of the following: 1) decrease in mPAP to ≤ 40 mm Hg; 2) decrease of at least 10 mm Hg in mPAP; and 3) unchanged or increased cardiac output. Choice A fits the definition of vasoreactivity. In Choice B, the mPAP does not fall enough, even though the cardiac output increases.

In Choice C, mPAP falls, but this may be because flow (cardiac output) falls significantly as well.

Choice D is incorrect, all patients with newly diagnosed PAH considered idiopathic, heritable or anorexigen-associated should have vasoreactivity testing.

Note that a subsequent research has demonstrated that vasoreactivity should only be tested in those with idiopathic, heritable, or anorexigen-associated PAH benefit, because those are the only three groups that, if vasoreactive, respond to first-line CCB treatment.² That same follow-up study showed that other sub-groups of PH, including connective tissue disorder-associated Group I PAH, and other World Health Organization (WHO) Groups of PH (Groups 2-5), do not benefit from CCB therapy, even if vasoreactive.

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